MIL-P-51406A(EA)
26 March 1984
SUPERSEDING
MIL-P-51406(EA)
29 October 1974

MILITARY SPECIFICATION

PLASTIC SHEETS, VINYL CHLORIDE POLYMER AND COPOLYMER, FLEXIBLE

This specification is approved for use by US Army Armament, Munitions and Chemical Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers transparent, flexible sheets, made of vinyl chloride polymer or copolymer resins.

1.2 Classification. The sheets shall be furnished in nominal thicknesses of 0.020, 0.040, and 0.060 inches, as specified (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-F-51110 - Faceblank - Lens, C14R3.

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, Armament Research and Development Center, US Army Armament, Munitions and Chemical Command, ATTN: DRSMC-TSC-S(A), Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter. :

FSC 9330

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STANDARDS

FEDERAL

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).
Fed. Std. No. 406 - Plastic, Methods of Testing.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-282 - Filter Units, Protective Clothing, Gas-Mask Components and Related Products: Performance-Test Methods.
MIL-STD-810 - Environmental Test Methods.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

NATIONAL BUREAU OF STANDARDS

Circular 478 - Colorimetry.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

ASTM STANDARDS

D568 - Test for Rate of Burning and/or Extent and Time of Burning of Flexible Plastics in a Vertical Position.
D570 - Test for Water Absorption of Plastics.
D635 - Test for Rate of Burning and/or Extent and Time of Burning of Self-Supporting Plastics in a Horizontal Position.
D638 - Test for Tensile Properties of Plastics.
D673 - Test for Mar Resistance of Plastics.
D746 - Test for Brittleness Temperature of Plastics and Elastomers by Impact.
D747 - Test for Stiffness of Plastics by Means of a Cantilevel Beam.

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D1003 - Test for Haze and Luminous Transmittance of Transparent Plastics.
D1203 - Test for Volatile Loss from Plastic Materials.
D1925 - Yellowness Index of Plastics.

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Materials. The sheets shall be made of virgin vinyl chloride polymer or copolymer resins and whatever plasticizers, stabilizers, and lubricants that might be required to enable the finished product to satisfy the requirements of this specification.

3.2 Thickness. The thickness of the sheets shall be within ± 10 percent of the nominal value (see 1.2).

3.3 Optical defects. Optical defects are defined as imbedded particles, scratches, bubbles, or any other imperfections which reduce visibility through the sheet. Excluding a 1-inch margin around the periphery, the aggregate of all defective areas shall not exceed 1 percent of the area of the sheet; furthermore, the sheet shall not contain more than 8 optical defects whose major dimensions exceeds 1/32 inch.

3.4 Physical and chemical properties. The sheets shall comply with the requirements in table I when tested in accordance with the applicable test method.

3.5 Workmanship. The sheets shall be colorless, smooth, and free from blemishes, straie, waves, and damage such as cracks, splits, and other visible defects which will impair their appearance or cause distortion of vision. In addition, the sheets shall be free from contamination.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Contractor's responsibility. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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TABLE I. Requirements and tests.

Property	Units	Requirement				Test method		
		Unaged		Aged (see 4.2.4.6)		Fed. Std. or ASTM	Method or para	Number of determination
		Min	Max	Min	Max			
Brittle temperature	°F		*		*	ASTM D 746		2
Crease resistance	degree	120					4.2.4.1	10
Discoloration	%		1		5		4.2.4.2	2
Flammability		Self-extinguishing				ASTM D 635 ASTM D 568		1
GB resistance	hour	4					4.2.4.3	**
HD resistance (vapor)	hour	4					4.2.4.4	**
Haze	%		4		4	ASTM D 1003		2
Optical uniformity	(angular) minutes		10			Fed. Std. No. 406	Method 3041	1
Mar resistance	%***		15			ASTM D 673		2
Stiffness in flexure at 75°F	psi		10,000		10,000	ASTM D 747		2
Tensile strength	psi	3,000		3,000		ASTM D 638		2
Ultimate elongation	%	250		100		ASTM D 638		3
Volatility of plasticizer	%		1.5			ASTM D1203		10
Water absorption at 75°F	%		0.25			ASTM D570		3
Water absorption at 212°F	%		0.45				4.2.4.5	3
Fungus		pass				MIL-STD-810	Method 508.1	2****
Weathering	Yellowness index (Non dimensional)		16 *****		16 *****		4.2.4.7	5

*Lower than -40°F for 0.020 inch sheet; lower than -10°F for thicker sheets.
**Each test specimen shall meet the minimum requirement. Results shall be reported to the nearest minute.
***Percent haze, according to ASTM D1003, using 400 grams of No. 80 silicon carbide.
****Each test specimen, 12 inches by 12 inches in size, shall show no evidence of interlaminar fungal growth.
*****Average yellowness value shall be no more than 14; no individual sample shall be greater than 16.

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4.1.2 Government's responsibility. The Government will be responsible for the performance of the tests in 4.2.4.3, 4.2.4.4, and 4.2.4.5. Samples shall be forwarded to the laboratory designated by the contracting officer.

4.1.3 Objective evidence. The contractor shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5, for which specific inspection has not been provided in this specification, have been satisfied.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of sheets of one thickness produced from one batch (see 6.3) of resin, under essentially the same conditions and offered for shipment at one time.

4.2.2 Sampling.

4.2.2.1 For nondestructive examination. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.2.2 For destructive tests. A representative sample, 30 square feet minimum, shall be taken from each lot to supply the specimens for the tests specified in table I.

4.2.2.3 For GB and HD resistance. A representative sample, 10 square feet minimum, shall be taken from each lot and submitted to the Government for GB and HD tests (see 4.1.2).

4.2.3 Inspection procedure.

4.2.3.1 Examination. Sample sheets from each lot shall be examined in accordance with the classification of defects and MIL-STD-105.

4.2.3.2 Destructive tests. All samples shall be tested in accordance with table I.

4.2.3.3 GB and HD resistance. Samples shall be forwarded for tests as outlined in 4.2.4.3 and 4.2.4.4, to the laboratory designated by the contracting officer (see 6.2).

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4.2.3.4 Classification of defects.

(a) Plastic sheets, vinyl chloride polymer and copolymer, flexible.

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Optical defects (3.3)
102	Thickness incorrect (3.2)
Minor:	AQL 2.5 percent defective
201	Damage (3.5)
202	Contamination (grease or oil)

4.2.4 Tests. (See table 1; Note: When the same value of a requirement is to be met by sheets in both aged and unaged conditions, one-half of the number of determinations shall be made in each condition. When more than one determination of a property in a specified condition is made, the results shall be averaged before comparison with the requirement, unless otherwise noted.)

4.2.4.1 Crease resistance. The crease resistance of the sheet shall be determined as follows, using specimens measuring 1 by 2 inches:

(a) Condition the specimens for 48 hours in an atmosphere maintained at a temperature of 75° ± 2°F and a relative humdity of 50 ± 2 percent.

(b) Fold each specimen to form a 1 inch square, and subject it to a uniform load of 25 pounds for 5 minutes.

(c) Remove the load and allow a 3 minute interval for free recovery; then immediately measure the dihedral angle formed at the crease. In the case of a 0.020 inch thick sheet, if the plies cohere when the load is removed, postpone the start of the 3 minute interval of free recovery until the instant when such cohesion is overcome. Do not delay, however, longer than 150 seconds.

4.2.4.2 Discoloration. The discoloration of the sheet shall be determined as follows:

(a) Irradiate the specimen with light conforming to standard illuminate C of the International Commission of Illumination (C.I.E.).

(b) By spectrophotometric analysis, in accordance with Circular 478, determine the excitation purity of the transmitted light relative to that of the incident light; disregard the dominant (or complimentary) wave length.

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(c) The deviation of excitation purity found in "b" is the percent of discoloration.

4.2.4.3 GB resistance (vapor). GB resistance shall be determined in accordance with method T208 of MIL-STD-282.

4.2.4.4 HD resistance (vapor). HD vapor resistance shall be determined in accordance with method T209 of MIL-STD-282.

4.2.4.5 Water absorption at 212°F. The water absorption of the sheet at 212°F shall be determined in accordance with the applicable portions of ASTM D570, except that boiling water shall be used in lieu of water maintained at 75°F, and the period of immersion in the boiling water shall be 1 hour.

4.2.4.6 Aging. The specimens shall be conditioned for one calendar week in circulating air maintained at 160° ± 2°F and a relative humidity of 50 ± 2 percent.

4.2.4.7 Weathering. Specimens, one square foot each, shall be taken from each lot to be subjected for 72 ± 2 hours using an atlas weatherometer (type x-w) operating without the sprayer and using the light cam only and tested in accordance with ASTM 1925.

4.2.5 Acceptance/rejection criteria. If any specimen or average of determinations, as applicable, fails to meet the applicable requirements of this specification, the lot represented shall be rejected.

5. PACKAGING

5.1 Packaging, packing and marking, interplant shipment (see 6.4). The vinyl chloride sheets shall be packaged and packed to provide adequate protection from physical damage from the supply source to the first receiving activity for immediate use or further processing. Shipping containers shall be in compliance with the rules and regulations applicable to the mode of transportation. Marking shall be in conformance with Fed. Std. No. 123 and shall include the contractor's name, address, and contract number.

6. NOTES

6.1 Intended use. The material covered by this specification is intended for use in the manufacture of chemical-biological mask lenses and related items.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) The sheet thickness required (see 1.2).
(c) The laboratory to which samples should be sent for GB and HD testing (see 4.2.3.3).

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6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Interplant shipment. Packaging for supplies and materials which will not enter the military supply system. Typical interplant shipments are shipments from a vendor to a subcontractor or a prime contractor, or from a subcontractor to a prime contractor, or from a vendor or contractor to a military arsenal or plant.

6.5 International interest. Certain provisions of this specification are the subject of international standardization agreement QSTAG-216 and 234. When amendment, revision, or cancellation of this specification is proposed which will affect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels including departmental standardization offices, if required.

Custodian:

Army - EA

Preparing activity:

Army - EA

Project No. 9330-A048

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effective. If the PCB or HUB teams need more

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-51406A(EA)	PLASTIC SHEETS, VINYL CHLORIDE POLYMER AND COPOLYMER, FLEXIBLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* - Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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